

PreLab Homework - 10

First task: Students Grades Project (60 pts.)

You will write a C program for this homework in all tasks. An input.txt file containing the names and grades of the students will be given to you. The number of students will be between 3 and 30. You will read this file and use the data in your program. Then you are expected to perform the operations listed in the menu below. The grade is between 0 and 100. After reading the input file, the program displays a menu which is shown below. The menu consists of 5 options:

1. Show most successful student,
2. Show most unsuccessful student,
3. List the number of students for each letter grade,
4. Show calculated average grade,
5. Show all four options data.
0. Quit

Options should give same results as in the figures and letter grades must be calculated programmatically according to grade scale which is shown in table below. The program will keep running until the user enters 0 in the menu. After entering 0, the program will terminate. Also, the menu is shown after every selection.

Note: You must use switch cases to construct the menu.

Grade	Letter Grade
90-100	A
80-89	B
70-79	C
60-69	D
0-59	F

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Student Score Calculator Menu for 30 Student
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1) Most Successful Student
2) Most Unsuccessful Student
3) Letter Grade Statistics
4) Calculate Average
5) Show all Data
0) Quit
Make Selection: 0

Quit selected!
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Student Score Calculator Menu for 30 Student

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- 1) Most Successful Student
- 2) Most Unsuccessful Student
- 3) Letter Grade Statistics
- 4) Calculate Average
- 5) Show all Data
- 0) Quit

Make Selection: 1

Most Successful Student:

Score: 98

Letter Grade: A

Swap Count: 214

Student: Kaan

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Student Score Calculator Menu for 30 Student

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- 1) Most Successful Student
- 2) Most Unsuccessful Student
- 3) Letter Grade Statistics
- 4) Calculate Average
- 5) Show all Data
- 0) Quit

Make Selection: 2

Most Unsuccessful Student:

Score: 11

Letter Grade: F

Swap Count: 220

Student: Tolga

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Student Score Calculator Menu for 30 Student

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- 1) Most Successful Student
- 2) Most Unsuccessful Student
- 3) Letter Grade Statistics
- 4) Calculate Average
- 5) Show all Data
- 0) Quit

Make Selection: 5

Most Successful Student

Score: 98

Letter Grade: A

Swap Count: 214

Student: Kaan

Most Unsuccessful Student

Score: 11

Letter Grade: F

Swap Count: 220

Student: Tolga

5 Student got letter grade 'A'
3 Student got letter grade 'B'
3 Student got letter grade 'C'
5 Student got letter grade 'D'
18 Student got letter grade 'F'

The avarage Score of 30 Student is 43.17

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Student Score Calculator Menu for 30 Student

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- 1) Most Successful Student
- 2) Most Unsuccessful Student
- 3) Letter Grade Statistics
- 4) Calculate Average
- 5) Show all Data
- 0) Quit

Make Selection: 3

5 Student got letter grade 'A'
3 Student got letter grade 'B'
3 Student got letter grade 'C'
5 Student got letter grade 'D'
18 Student got letter grade 'F'

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Student Score Calculator Menu for 30 Student

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- 1) Most Successful Student
- 2) Most Unsuccessful Student
- 3) Letter Grade Statistics
- 4) Calculate Average
- 5) Show all Data
- 0) Quit

Make Selection: 4

The avarage Score of 30 Student is 43.17

While implementing options 1 and 2, sorting algorithms must be used.

For option 1, write the Bubble Sort algorithm; for option 2, write the Quick Sort algorithm.

During the execution of both algorithms, calculate how many swap operations occur while sorting the array. Print this number on the screen as "Swap count: ".

Sort the randomly generated array in descending order using Bubble Sort.

Sort the randomly generated array in ascending order using Quick Sort.

Note: There will be no duplicate scores in the input.txt file; in other words, no two students will have the same grade. Grades in the input file will be integer and they are random numbers between 0-100, including both 0 and 100 grades. The average value may be a decimal number. Therefore, you must print the average with two digits after the decimal point (for example: not print 73.45123, print 73.45).

Second Task: Sorting Performance Comparison (40 pts.)

You are required to compare the performance of two different sorting algorithms (Bubble Sort and Quick Sort) by measuring and analyzing their execution times.

1) You will write a C program that:

Generates arrays of different sizes (n),

where n will be values as: 10000, 20000, 30000, 200000, 300000, 500000

2) Populates each array with random integers between 0 and length of the array.

3) Sorts the arrays using two different sorting algorithms: Bubble Sort and Quick Sort.

4) Measures the execution time for each algorithm on each array size. You will use the clock() function from <time.h> to record time duration.

5) Compare the performance of the two algorithms based on time in ms.

6) Manually plot the graphs of the algorithms' execution times on paper, then take a photo of your drawing and include it in your assignment to compare their performance.

Good Luck!